

Project 1: Data Cleaning & Preparation

DecodeLabs Industrial Training Kit

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1. Objective

The objective of this project was to clean and prepare a raw dataset for analysis by identifying missing values, handling incomplete data, verifying duplicate records, standardizing data formats, and validating business rules.

2. Dataset Overview

The dataset contains 1,200 records and 14 columns representing customer orders, including Order ID, Date, Customer ID, Product, Quantity, Unit Price, Payment Method, Coupon Code, Referral Source, and Total Price.

3. Data Cleaning Tasks Performed

- Identified missing values using Pandas.
- Replaced missing values in the CouponCode column with 'No Coupon'.
- Verified that no duplicate rows existed.
- Verified that all Order IDs were unique.
- Standardized the Date column to datetime format.
- Removed leading and trailing spaces from text columns.
- Standardized categorical text using Title Case.
- Validated TotalPrice using $\text{Quantity} \times \text{UnitPrice}$ with rounded comparison to avoid floating-point precision issues.

4. Final Validation Results

After cleaning and validation:

- Missing Values: 0
- Duplicate Rows: 0
- Duplicate Order IDs: 0
- Date Format: Standardized (datetime64)
- Business Logic Validation: Passed

5. Conclusion

The dataset is clean, consistent, and ready for exploratory data analysis, dashboard development, or machine learning tasks. All required data cleaning and validation steps from Project 1 were completed successfully.

Appendix: Change Log

Change ID	Description	Status
CR001	Filled missing CouponCode values with 'No Coupon'	Completed

CR002	Verified duplicate rows	Completed
CR003	Verified unique Order IDs	Completed
CR004	Standardized Date format	Completed
CR005	Trimmed text columns	Completed
CR006	Standardized text capitalization	Completed
CR007	Validated TotalPrice business rule	Completed